Title: Study of Amplitude Modulator and Demodulator using MATLAB

Objectives:

The main objective of this experiment was to study amplitude modulation and demodulation using MATLAB simulation. Specially, the experiment aimed to understand the principle of AM, design and analyze it's block diagram and observe the signals involved at different stages of modulation and demodulation. Another important goal was to analyze the impact of modulation and to ensure the recovery of the original message signal during demodulation.

Theory:

AM Modulation:

Amplitude modulation (AM) is a one of the conventional techniques used to transmit message signals using a carrier wave. The amplitude or strength of the high frequency carrier wave is modified in accordance with amplitude of the message signal.

Let us consider $m(t)$ is the message signal, and $c(t)$ is the high frequency carrier signal:

$m(t) = A_m\sin(\omega_m t) = Am\sin(2\pi f_m t)$

$c(t) = A_c\sin(\omega_c t) = Ac\ \sin(2\pi f_c t)$

Where,

- A_c= Carrier signal amplitude
- A_m = Message signal amplitude
- f_c= Carrier frequency
- f_m =Message frequency

Then, AM modulated signal can be constructed as follows:

$am(t) = [Ac + m(t)]\ \sin(\omega_c t)$

$= [A_c + A_m\sin(\omega m t)]\ \sin(\omega_c t)$

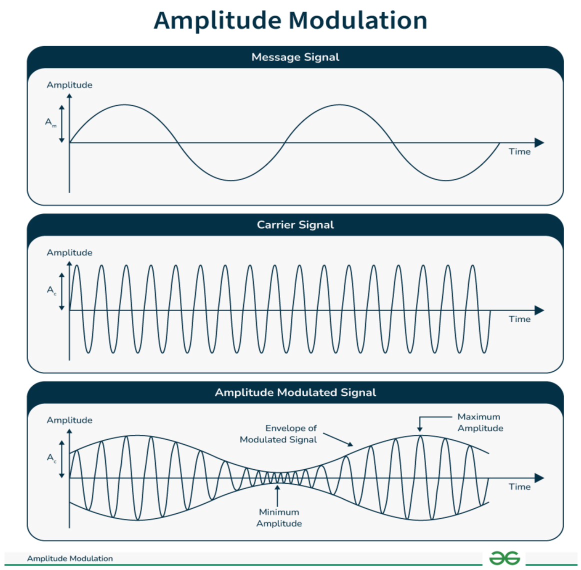
$= [A_c\ \{1+\frac{A_m}{A_c}\ \sin(\omega_m t)\}]\ \sin(\omega_c t)$

$= [A_c\ \{1 + \mu\sin(\omega_m t)\}]\ \sin(\omega_c t)$

$=A_c\sin(\omega_c t) + \mu Ac\ \sin(\omega_c t)\sin(\omega_m t)$

$=A_c\sin(\omega_c t) + \frac{\mu Ac}{2}\ [\cos\{(\omega_c - \omega_m)t\} - \cos\{(\omega_c + \omega_m)t\}]$

Here, $\mu = \frac{A_m}{A_c}$ known as AM modulation index. The value of $0 \leq \mu \leq 1$. Thus, $Am < Ac$.



AM Demodulation:

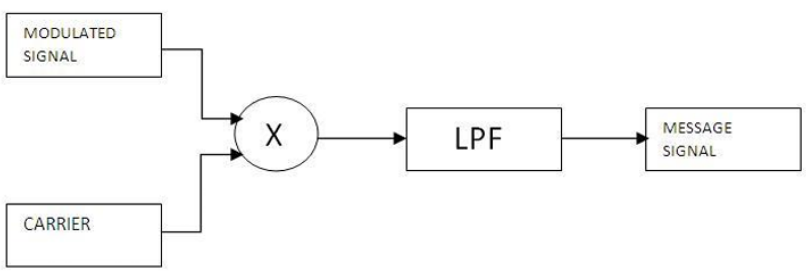


Fig. 2. AM demodulation process.

Demodulation methods in amplitude modulation (AM) are techniques employed to extract the original modulating signal (such as audio and speech) from the modulated carrier wave. Thus, if we multiply the AM modulated signal with the carrier signal again in the demodulation process, one component becomes higher frequency $2\omega_c$ and other component becomes the low frequency baseband/modulating signal frequency ω_m as shown below:

$$A_c \cos\{(\omega_c - \omega_m)t\}\sin(\omega_c t)$$

$$= \frac{A_c}{2}[\sin\{(\omega_c - \omega_m + \omega_c)t\} - \sin\{(\omega c - \omega m - \omega c)t\}]$$

$$= \frac{A_c}{2}[\sin\{(\boldsymbol{2\omega_c} - \omega_m)t\} + \sin\{(\boldsymbol{\omega_m})t\}$$

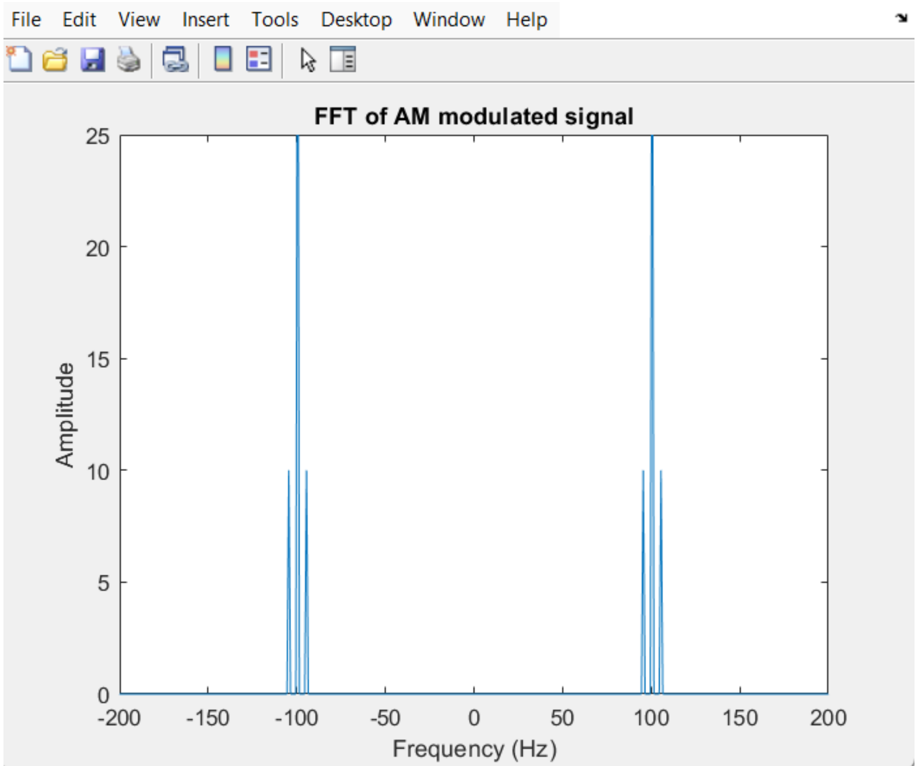
Code:

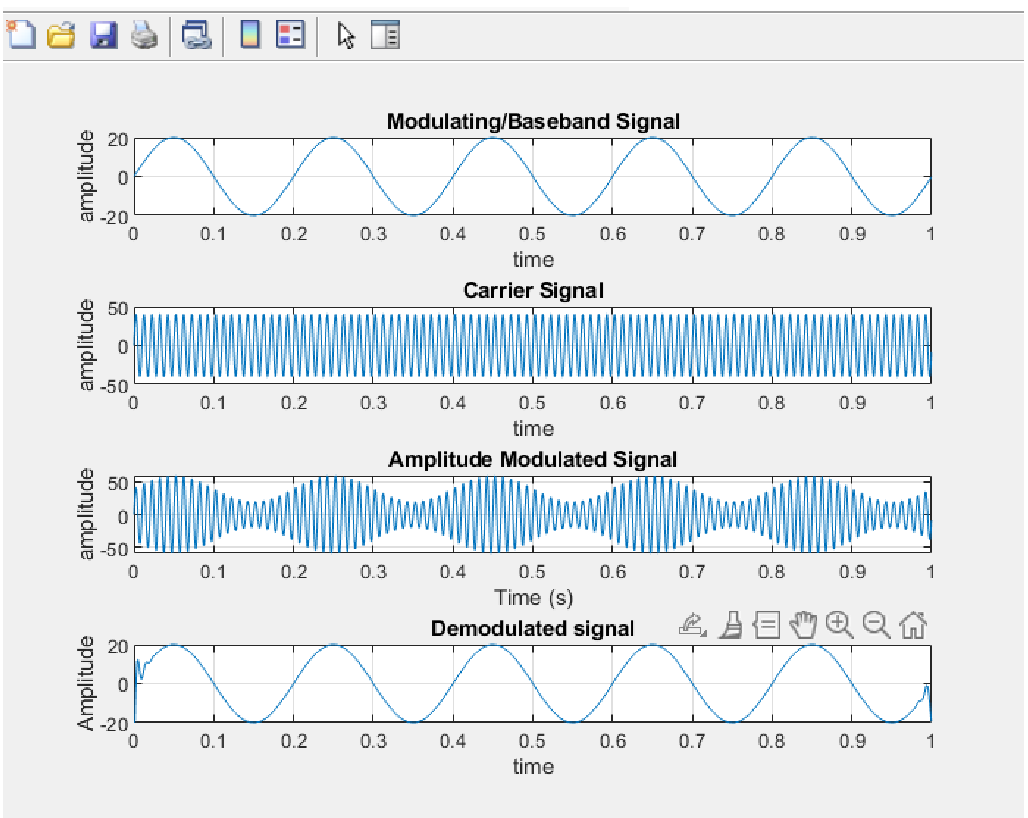
```
clc;
clear;
%baseband signal,  m(t) = 20sin(10*pi*t)

fs=3000;
t = 0:1/fs:1-1/fs;
fm=5;
fc=100;

% Baseband Signal (Modulating Signal)
Am=20; % amplitude of baseband signal
m = Am.*sin(2*pi*fm*t);

% Carrier Signal
Ac=40; % amplitude of carrier singal
c= Ac.*sin(2*pi*fc*t);

% modulation index
mu=Am/Ac; % mu=10/20=0.5

% AM Modulated Signal
cam=(Ac.*(1+mu*sin(2*pi*fm*t))).*sin(2*pi*fc*t);

fftSignal = fft(cam); % This is frequency response of amplitude modulated signal (cam).
fftSignal = fftshift(fftSignal)/(fs/2);
f = fs/2*linspace(-1,1,fs);
```

```
% AM demodulatation part

%multiplying the am modulated signal with carrier signal
am_demodulated =cam.*c;
%Applying Low-Pass filter
[k,l] = butter(6,(100).*2/fs);
filtered_signal = filtfilt(k,l,(am_demodulated-800)./20) ;

% Plot the signals
figure;
subplot(4,1,1);
plot(t,m);
ylabel('amplitude');xlabel('time');
title('Modulating/Baseband Signal');
grid on

subplot(4,1,2);
plot(t,c);
ylabel('amplitude');xlabel('time');
title('Carrier Signal');
grid on

subplot(4,1,3);
plot(t,cam);
ylabel('amplitude');xlabel('time');
title('Amplitude Modulated Signal');
xlabel('Time (s)');
grid on
```

```
subplot(4,1,4);
%plot(t,am_demodulated);
plot(t,filtered_signal);
ylabel('Amplitude');xlabel('time');
title('Demodulated signal');
axis ([0 1 -20 20]);
grid on

figure;
plot(f, abs(fftSignal));
axis([-200 200 0 25]);
title('FFT of AM modulated signal');
xlabel('Frequency (Hz)');
ylabel('Amplitude');
```

Simulation figure:





Task 1. Define modulation and AM modulation. Why is modulation necessary in communication?

Ans:

Modulation is the process of varying the property of a high-frequency carrier signal (like amplitude, frequency, or phase) according to the information/message signal. Amplitude Modulation (AM) is a type of modulation where the amplitude of the carrier wave changes in proportion to the instantaneous amplitude of the message signal. Modulation is necessary because message signals like voice or data usually have low frequency and cannot travel long distances effectively. Modulation shifts the signal to a higher frequency range making it easier to transmit over long distances. It reduces antenna size requirements, allows multiple signals to share the same medium and improves signal quality by reducing noise and interference.

Task 2. Mathematically proved that for the message signal $m(t)$ = A_m sin($\omega_m t$) and carrier signal $c(t)$ = c sin($\omega_c t$), AM modulated signal is a composite signal consisting of three frequency components. Then, produce the equation for Bandwidth for AM modulated signal.

Ans:

Given:

- Message signal: $m(t) = A_m \sin(\omega_m t)$
- Carrier signal: $c(t) = A_c \sin(\omega_c t)$

The AM modulated signal is:

$$\begin{aligned} s(t) &= \{A_c + m(t)\}\sin(\omega_c t) \\ &= \{A_c + A_m \sin(\omega_m t)\}\sin(\omega_c t) \end{aligned}$$

Expanding:

$$s(t) = A_c \sin(\omega_c t) + \frac{A_m}{2}[\cos\{(\omega_c - \omega_m)t\} - \cos\{(\omega_c + \omega_m)t\}]$$

This shows the AM signal consists of three frequency components:

1. Carrier frequency (ω_c)
2. Upper sideband ($\omega_c + \omega_m$)
3. Lower sideband ($\omega_c - \omega_m$)

Bandwidth (BW):

$$BW = 2f_m$$

where f_m is the frequency of the message signal.
So, the AM signal requires twice the message frequency as bandwidth.

Task 3. Write a MATLAB code to generate AM modulated and demodulated signal for baseband signal m(t) = 20sin($10\pi t$).
Ans: Given in the code

Result analysis/discussion:

In this experiment, the MATLAB simulation showed how amplitude modulation works in practice. The carrier signal's strength changed according to the message signal which means the information was successfully carried by the wave. When demodulation was applied the original message was recovered almost the same as before proving that the process works well. Some small distortions were noticed depending on the chosen frequency and filter settings but overall the results matched the theory. This makes it clear that amplitude modulation and demodulation are effective ways to send and receive information and MATLAB helped visualize the whole process in a simple and clear way.

Conclusion:

The experiment successfully demonstrated the principles of amplitude modulation and demodulation using MATLAB simulation. The original message signal was recovered from the modulated waveform, validating the theoretical concepts studied in communication systems. The analysis of the modulation index emphasized its critical role in ensuring signals where modulation and over modulation both reduce efficiency and clarity.

References:

[1] MathWorks, MATLAB Documentation, Natick, MA, USA: MathWorks. [Online]. Available: https://www.mathworks.com/help/

[2] Stormy Attaway, MATLAB: A Practical Introduction to Programming and Problem Solving, 6th ed. Boston, MA, USA: Butterworth-Heinemann, 2019